

Michelle Egan and Dimitris Tsarouhas (eds.), *Europe in a Weaponized World*, chapter proposal --Juljan Krause, INTA, Georgia Tech

Qubits, Sovereignty, and Strategic Ambitions: Europe's Quantum Technologies in a Weaponized World

Abstract:

Quantum technologies occupy an increasingly important place in Europe's evolving security posture and the bloc's strategic ambitions for greater technological sovereignty amid a rapidly shifting alliance structure. Long associated with fundamental research and future-oriented innovation largely unaffected by geostrategic concerns, quantum technologies (here defined as encompassing quantum computing, quantum communication, and quantum sensing), are increasingly framed as vital strategic assets with considerable implications for cybersecurity, defense, industrial competitiveness, and technological independence. National and EU-level quantum strategies vividly reflect a new politics of strategic competition, in which frontier technologies such as quantum, though largely aspirational at present, are expected to deliver dramatic and sustained competitive advantage in the near term. This chapter traces how Europe's quantum effort has evolved, and asks what the European quantum agenda reveals about the Union's broader adaptation to a weaponized international environment.

It argues that quantum technologies illustrate a wider transformation in European policy, one that seeks to reposition emerging-technology policy away from a predominantly market-enabling model of innovation governance towards a more security-conscious model of technological statecraft. European quantum initiatives aim to build domestic capabilities, protect sensitive information, reduce strategic dependencies, and position Europe within global competition over the critical infrastructures of the future. Quantum technologies matter for Europe not only for their technical capabilities, but because they allow Europe to signal ambitions for quantum advantage long before any such capacities are realized. In quantum policy, Europe's concerns over access to future-forming technologies, industrial competitiveness, defense innovation, and critical infrastructure resilience crystallize.

Yet Europe's quantum posture remains marked by a persistent tension between openness and control, collaboration and protection, and strategic ambition and institutional fragmentation. EU-level initiatives compete with the national strategies of influential member states, and the Union's ambitions remain embedded in transatlantic research networks, global supply chains, and private-sector platforms, and dependent on uneven national capabilities. Highlighting the many obstacles on the path towards European quantum advantage, the chapter complicates the common framing of quantum as a straightforward route to autonomy. Quantum Europe, in this sense, is both an aspiration and a diagnostic as it reveals the possibilities, contradictions, and limits of European technological sovereignty in a world increasingly marked by weaponization.